Bench bill of materials

Every physical object the session needs. Tick it, then start.

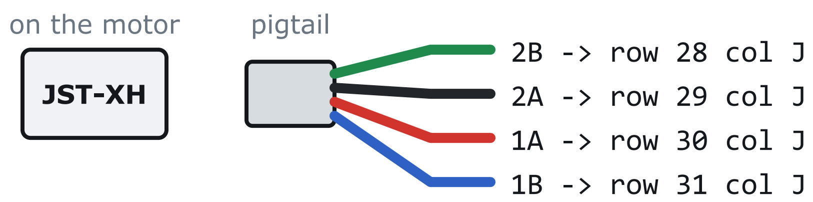
NOTHING HERE NEEDS ACQUIRING. Every item was confirmed on hand on 2026-09-11, the 830-point breadboard included. This box exists so that anything a later revision adds lands on page 1 rather than being discovered on page 13.

- [ ] **TMC2209 StepStick (FYSETC)** — 1 of 6
  The driver. Headers arrived PRE-SOLDERED, confirmed by opening the box, so page 5 is a skip. All six are one revision, so one Vref carries to all five later.
- [ ] **ESP32-C5-DevKitC-1** — 1
  Already on the bench. It stays OFF the breadboard - page 7 says why - so every wire that reaches it is male-to-female.
- [ ] **NEMA 17 + printed stand-in axle** — 1
  The load. The axle is PLA, which is why the firmware clamps every commanded speed to one drum revolution per second and why that cap is not configurable.
- [ ] **Breadboard, full-size 830-point** — 1
  63 rows, two rail pairs, an 0.3 in centre channel. The channel is the only reason a StepStick works on a breadboard at all - page 7.
- [ ] **Jumper wires, male-to-FEMALE** — 5
  Three signals (STEP, DIR, EN) and the two rail feeds. Female onto the ESP32's pins, male into the board.
- [ ] **Jumper wires, male-to-male** — 4
  Board-internal only: VIO, GND (logic), MS1 and MS2 to the rails. Both counts are derived from the parsed connection table, not typed on this page.
- [ ] **JST-XH 4-way mating pigtail** — 1
  THE MOTOR PLUG CANNOT ENTER A BREADBOARD. The pigtail's four flying leads can. Page 2 shows exactly where they terminate.
- [ ] **100 uF electrolytic, 35 V or more** — 1
  The bulk capacitor. Its legs need trimming and bending before it will reach the right two holes - lead prep on page 2.
- [ ] **Hookup wire, 22 AWG SOLID core** — 30 cm red + black
  RotoPD screw terminal to the board. Solid, not stranded: stranded frays in a breadboard hole, will not hold, and leaves a strand behind.
- [ ] **Flat screwdriver, 2 mm blade** — 1
  Two jobs: the RotoPD's screw terminals, and the Vref trimpot on page 23. A blade that does not fit the pot slips off it onto a pad.
- [ ] **Multimeter with a continuity beep** — 1
  Coil pairs (page 8), continuity (page 22), Vref (page 23). Continuity and DC volts on a 2 V range are the only functions used.
- [ ] **USB-C cable, PC to ESP32** — 1
  Console and logic power. A DATA cable - a charge-only lead gives a board that powers up, looks alive, and never appears as a serial port.
- [ ] **USB-C PD cable + RotoPD trigger** — 1
  VM. 9 V while the wiring is being proven, then 20 V before the soak, from the same trigger board.
- [ ] **Hex key for the drum set screw** — 1
  Fit the stand-in drum, and MARK it - a tape flag will do - so that `step 0 3200` is a readable result rather than a guess.
- [ ] **Heatsink (supplied, loose)** — 0 - NOT this session
  Deliberately not installed. Page 2 has the reasoning and the one condition that would change it.
- [ ] **Hall sensor + magnet** — 0 - NOT this session
  Branch B: no Hall fitted, so no homing, no edge figures, no revs. That is the expected configuration for this session and not a fault.

What the schematic does not say

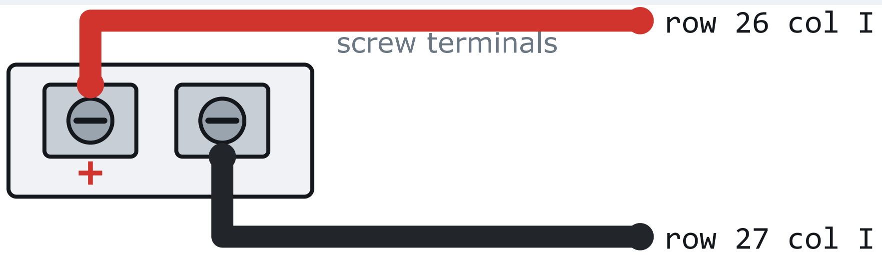
Three parts need preparing, one decision is already made, and one thing is deliberately NOT checked

1 The motor plug cannot enter a breadboard



The loom ends in a 4-way JST-XH. Its shell will not go into an 0.1 in board and forcing it spreads the contacts permanently. The mating pigtail plugs onto it and ends in four bare leads: strip 6 mm, twist each hard or tin it, and push them into rows 28 to 31 of column J. WHICH lead goes where is decided by the meter on page 8, never by colour - these wires were cut and re-terminated.

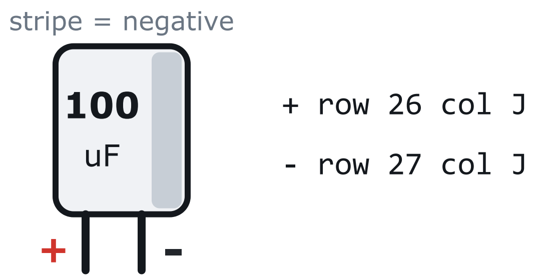
2 The RotoPD reaches the board on two wires



Strip 8 mm of the 22 AWG solid core, clamp it under the screw, and run the other end straight into the driver's own row. VM never goes through a rail.

BELIEVED to be screw terminals; the unit is unopened. If yours is not, STOP AND REPORT - do not improvise a connector on a 20 V rail.

3 The capacitor needs its legs prepared



It bridges the driver's OWN VM and GND, which are adjacent rows - 2.54 mm apart, against the 3.5 or 5 mm the leads came at. Trim both to about 12 mm, bend them toward each other at the body, and check the bends cannot touch. The body may overhang other rows; only the legs matter. Backwards, it vents.

4 Heatsink: not needed this session

DECIDED: the driver runs bare. 0.7 A RMS against the module's 2 A rating is roughly an eighth of rated dissipation, and the soak's thermal question is the motor sealed in PLA, not the driver. Page 28's STOP NOW list already carries 'the driver too hot to keep a finger on', so this is watched rather than assumed.

Not independently verified: the exact RDS(on)-derived junction temperature. IF one ever goes on, the adhesive pad MUST clear the Vref trimpot - a heatsink overhanging the pot ends page 23.

5 The current path, so nobody over-engineers

0.7 A RMS through 22 AWG wire, stock jumpers and breadboard rails is fine and needs no thought. A breadboard contact is good for about 1 A continuous and a jumper drops tens of millivolts here. Two things are worth KNOWING rather than fixing: the 0.7 A coil current flows in the four motor leads and inside the driver, never along a rail; and VM and GND come in on their own wires to the driver's own rows to keep the switching return out of the logic ground, not for current rating. The bulk capacitor, not the supply wire, delivers each step's transient. Upgrade nothing.

6 What the guard checks, and what it cannot

Every PIN on every page is parsed out of hal/pins.h and docs/BENCH_WIRING.md, cross-checked against the other, and the generator emits nothing at all if they disagree. That is why a drawing here cannot quietly contradict the firmware.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

It is the same distinction page 6 draws between a LABEL and a drawn position. The coordinates are chosen in tools/wiringgen.py, they are consistent with each other, and consistent is not verified.

Bench wiring — module V1

First power-on, connection by connection

The whole job in 28 pages, in the order you should do it. One connection per page. Every pin number here is read out of hal/pins.h and docs/BENCH_WIRING.md when these pages are rendered, so a drawing cannot quietly disagree with the firmware. Row and column numbers are a different kind of statement - page 2 says which.

The Hall sensor and magnet are NOT fitted. No homing, no position, no closed loop this round. That is expected, not a fault.

READ PAGE 4 FIRST. The three rules on it are how TMC2209 drivers die.

THE ORDER, and why it is this order

1-2	collect the parts, and prepare three of them
3-5	the rules, and the one solder job (skipped)
6-8	identify the parts, the board, the coil pairs
9-13	wire the logic (driver to ESP32)
14-18	wire the microsteps, then the power
19-21	the bulk capacitor, then the board as BUILT
22	**continuity check <- find mistakes with a meter**
23	**set Vref <- motor still NOT connected**
24-25	**connect the motor <- power off while you do it**
26-28	finished layout, the built board, power on / off

WIRE COLOURS — the same on every page

VM (motor +)	RED, thick
GND (power)	BLACK, thick
VIO -> 3V3	red, thin
GND (logic)	black, thin
STEP	orange
DIR	yellow
EN	white / grey
MS1	violet
MS2	grey-blue
PDN	brown

The motor keeps its own colours: red, blue, green, black.
Use any wire you like — but be consistent, and
never use red or black for a signal.

Three ways to kill the driver

Read before you pick anything up

1

NEVER connect or disconnect the motor while VM is live.

An open coil on a live driver dumps its stored energy into the output stage. This kills more StepSticks than every other cause put together. Power down FIRST, then touch the motor plug — every time, including 'just for a second'.

2

NEVER insert or remove the driver module while VM is live.

Same physics, and you will bridge something on the way past as well.

3

Set Vref BEFORE the motor is ever connected.

Vref sets the coil current. Set it with the motor attached and the first thing your motor sees is wherever the pot happened to be when it left the factory — often well over 1.5 A. This guide sets Vref on page 17 and connects the motor on pages 24-25, in that order, deliberately.

And one that is less dramatic but more common

The bulk capacitor must sit at the driver's OWN VM and GND pins — not on a breadboard rail at the far end of the board. Page 19 shows where.

If you are unsure at any point: power off, and nothing you do next can break anything. Every irreversible mistake in this guide needs VM to be live.

Solder the headers

SKIPPED this session - the modules arrived with headers already on

SKIP THIS PAGE AND GO TO PAGE 6. The FYSETC modules arrived PRE-SOLDERED - confirmed by opening the box, not inferred from a product photo. The page stays because the next module to arrive may not be, and because the orientation below is the one thing that cannot be undone.

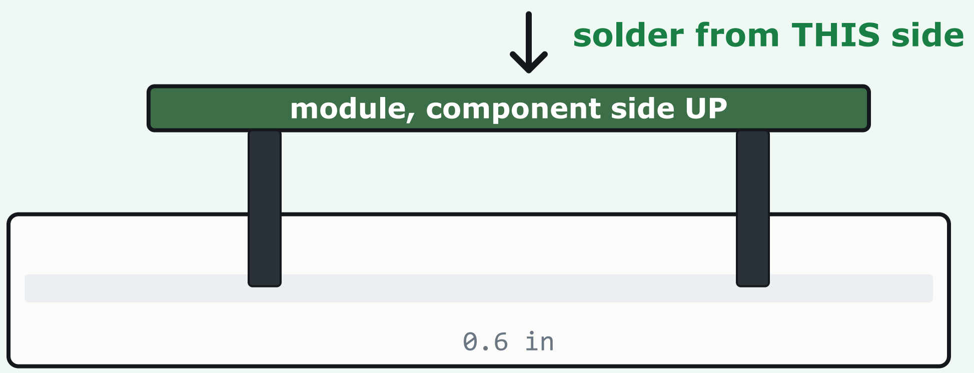
Confirm it in five seconds before you move on

- [ ] Two black strips of 8 pins each are IN the module, not loose in the bag.
- [ ] Every pin has a shiny cone of solder where it meets the board.
- [ ] Sighted from the end, all sixteen pins are parallel and the same length.
- [ ] The pins point AWAY from the components, i.e. out of the underside.

If a future module arrives with loose strips

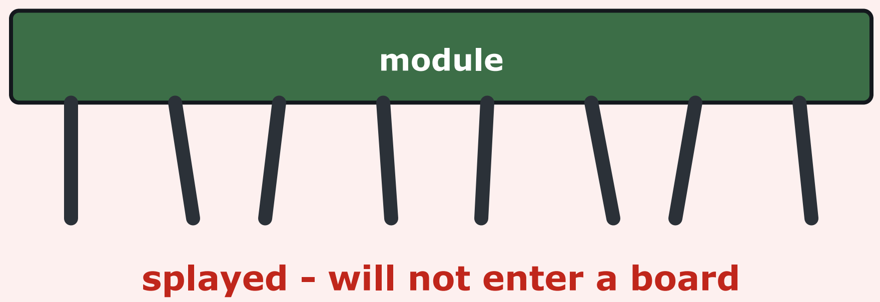
Solder it STRADDLED INTO THE BREADBOARD. The board holds all sixteen pins square and at the right spacing while you work, which is the whole problem - a module soldered freehand comes out with splayed pins that will not enter a breadboard, and straightening them afterwards breaks them at the joint.

RIGHT - pins held by the board



Strips into the board pins-down, module dropped on top, solder the sixteen joints you can see.

WRONG - pins held by hand or tape



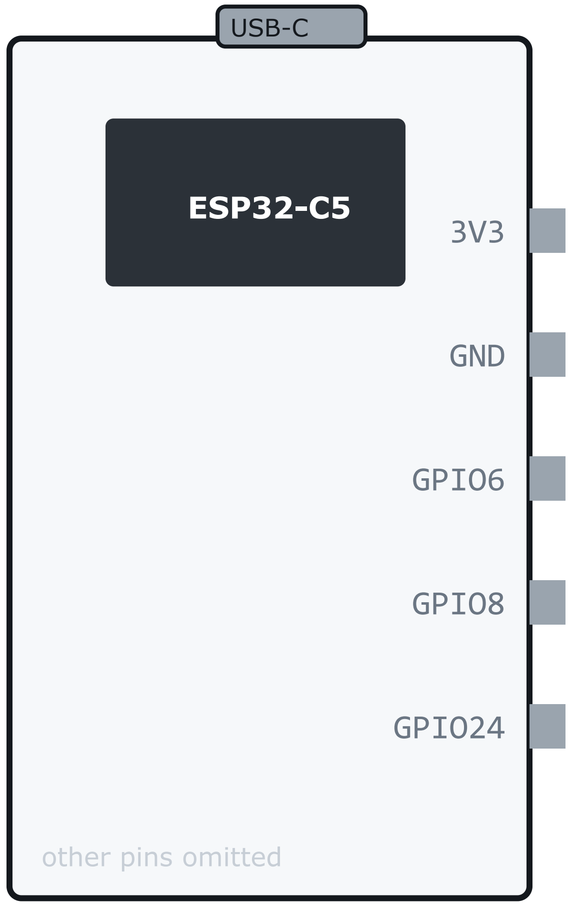
Sixteen pins cannot be held square by hand. This is the failure that costs a module rather than a minute.

Identify your two parts

Match by LABEL — drawn positions are only a hint

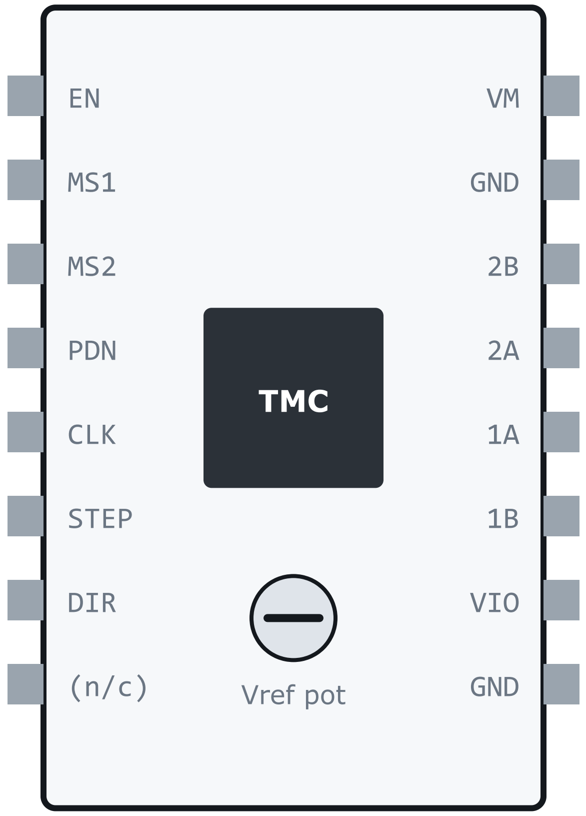
ESP32-C5-DevKitC-1

USB-C at the top



FYSETC TMC2209

standalone · match by LABEL



Pin ORDER here is the standard StepStick arrangement. If your module's silkscreen puts a label somewhere else, THE LABEL WINS. Every instruction in this guide names a pin and never a position.

Tick off these labels on the driver:

- [ ] EN
- [ ] MS1
- [ ] MS2
- [ ] PDN / UART
- [ ] STEP
- [ ] DIR
- [ ] VM / VMOT
- [ ] GND (x2)
- [ ] 1A
- [ ] 1B
- [ ] 2A
- [ ] 2B
- [ ] VIO / VDD

Find the two SENSE RESISTORS as well

Two small surface-mount parts beside the motor output pins, marked R110, R150 or R100. That marking decides your Vref target on page 23, and reading it takes ten seconds where guessing it costs you the whole thermal result.

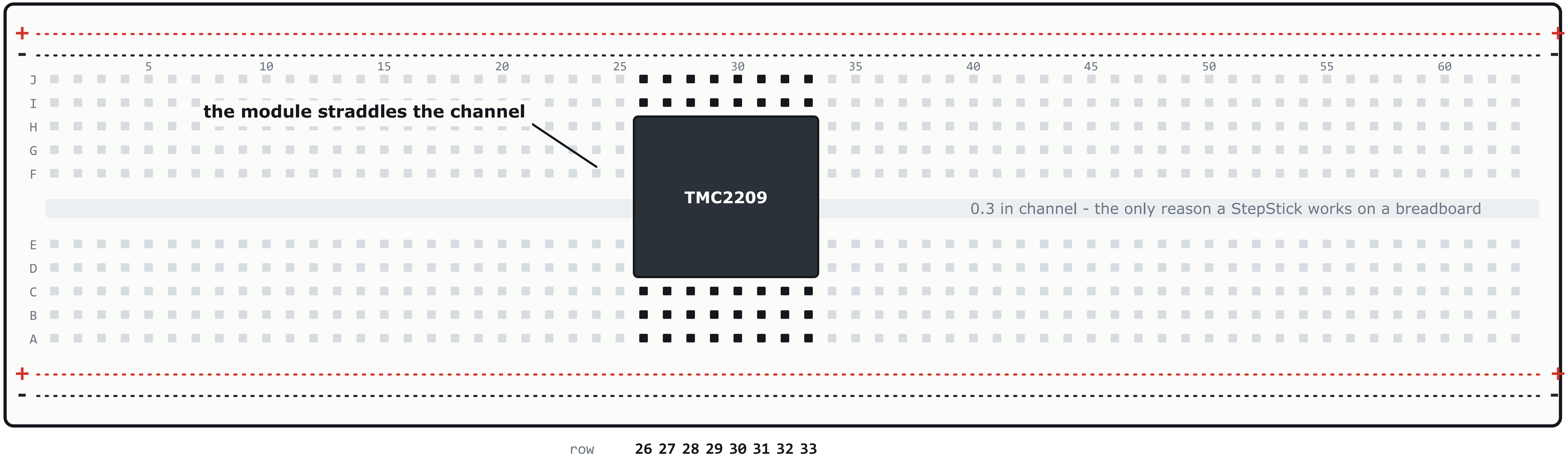
R _____ = 0. _____ ohm

R110 = 0.11 ohm · R150 = 0.15 ohm · R100 = 0.10 ohm

The breadboard, and where the driver sits

Nothing is wired yet. This is the board, drawn as it is.

Two facts about this board do all the work. Each group of five holes across ONE row, on one side of the channel, is ONE electrical node - so a wire in row 26 column J is electrically the driver's VM pin. And the four long strips along the edges are the POWER RAILS, which are joined to nothing at all until page 20 makes them live.



Why the driver lands in columns D and H

A StepStick's two headers are 0.6 in apart. Counting across the channel, column D to column H is 0.1+0.1+0.1+0.3+0.1+0.1 = 0.6 in exactly, so the module drops in with no pin sharing a node with its opposite number. That spacing is the one certain physical fact these coordinates are built on.

COLUMNS E, F AND G DO NOT EXIST for this build: the module body sits over them. Every wire on the right half goes into I or J, every wire on the left half into B. That is clearance, not tidiness.

THE ESP32 STAYS OFF THE BOARD

Its header-to-header spacing is 0.9 in or 1.0 in depending on the inset, and on an 0.1 in board with an 0.3 in channel both of them fit - landing the pins in different columns. Which GPIO sits where along the header is a board fact this repository does not carry, and guessing bends pins. So the DevKitC-1 lies loose on the bench and every wire to it is male-to-FEMALE.

Find these five by label on the silkscreen:

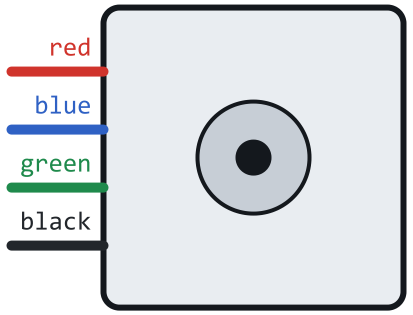
- [ ] GPIO8 (STEP)
- [ ] GPIO24 (DIR)
- [ ] GPIO6 (EN)
- [ ] 3V3
- [ ] GND

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

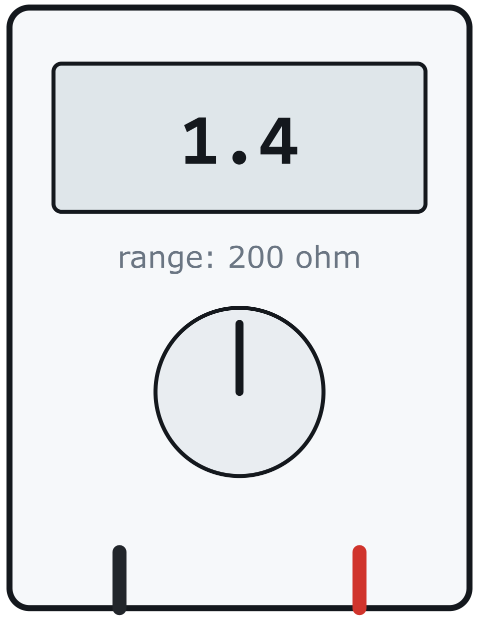
Find the coil pairs

Motor connected to NOTHING. Meter only.

LDO 42STH48-2504AH



DMM



The wires were cut and re-terminated, so no colour convention is evidence any more. Measure all six combinations. Exactly two read low — those are the coils. The other four must read open.

FIRST touch the probes together and note the reading (0.1-0.5 ohm). That is your test leads. Subtract it from everything below.

red - blue	______	ohm
red - green	______	ohm
red - black	______	ohm
blue - green	______	ohm
blue - black	______	ohm
green - black	______	ohm

Two low (about 1-2 ohm), four open. More than two low is a short — usually a stray strand on a Dupont pin. Any wire beeping to the motor CASE: stop, that winding is shorted to the frame.

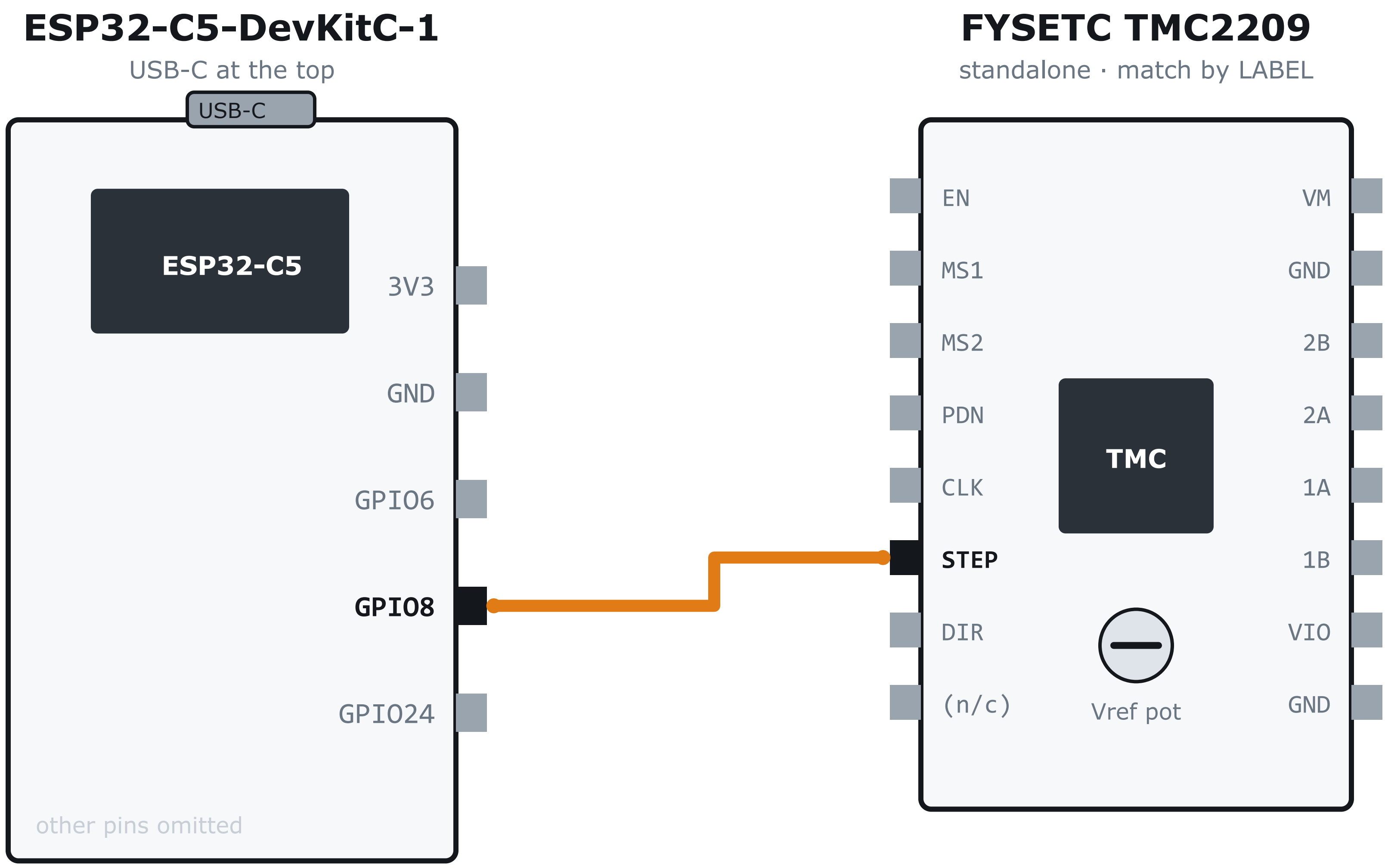
CROSS-CHECK WITH NO METER: twist one candidate pair's bare ends together and turn the shaft. A real pair makes it noticeably harder and notchier to turn; a wrong pair turns freely. Ten seconds, and it catches mistakes a meter reading does not.

Write it down — pages 24 and 25 refer to it:

PAIR 1 = ______________________________

PAIR 2 = ______________________________

Expected for this motor: black+green one coil, red+blue the other. IF YOUR METER DISAGREES, THE METER WINS.



One pulse on this wire moves the motor one microstep. At 1/16 microstepping that is 1/64th of a flap, and 3200 pulses is exactly one drum revolution. This is the only wire that carries motion.

ON THE BOARD

STEP is row 31 column D. Jumper (M-F) row 31 column B -> the ESP pin labelled GPIO8.

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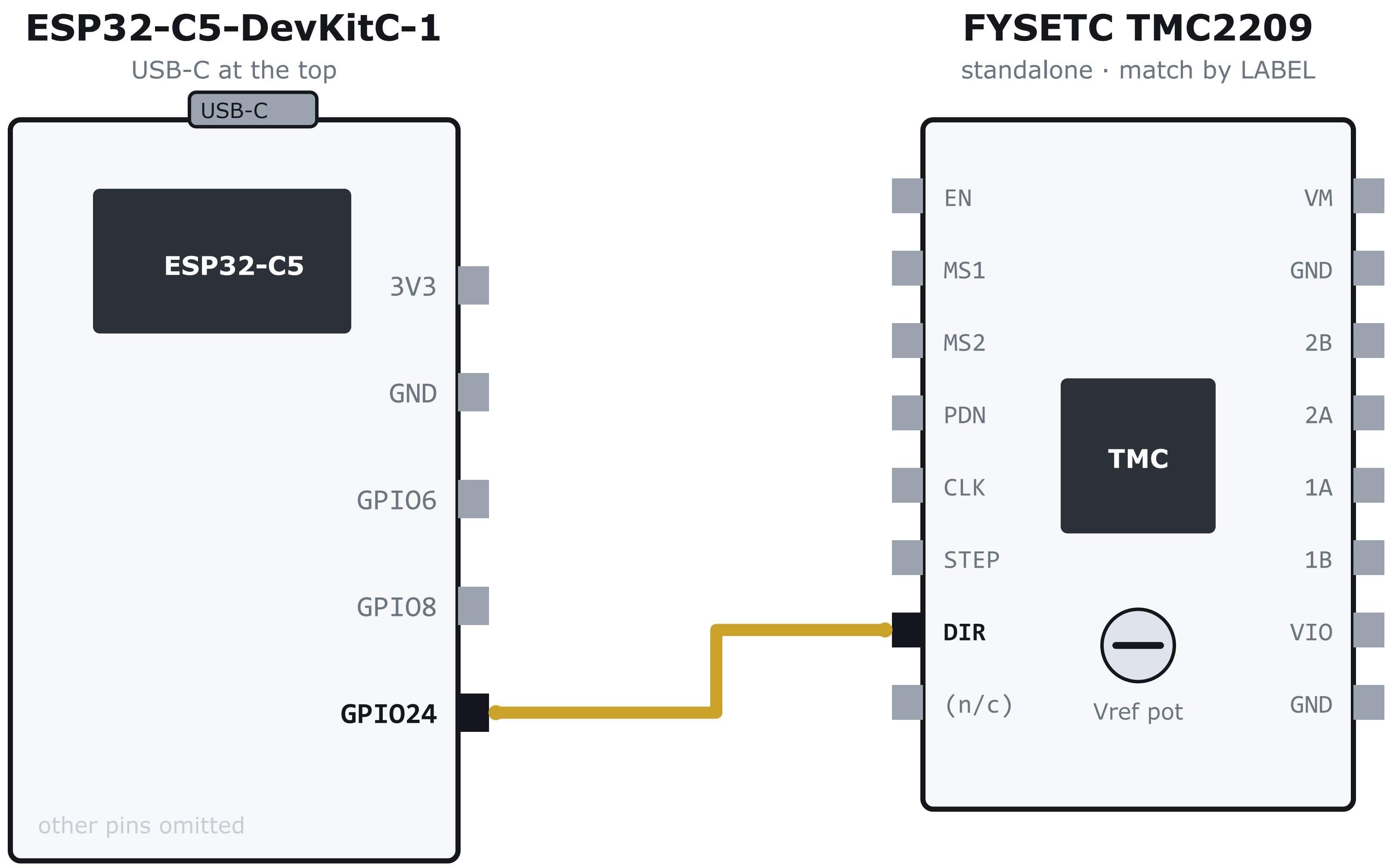
Connection 4 — DIR -> GPIO24

wire colour: yellow



A level, not a pulse: the driver reads it on each STEP edge to decide which way to turn. The ring is descending, so getting this backwards makes a countdown count up — which is why it is a firmware bit you can flip with `dir` rather than something to get right with a soldering iron.

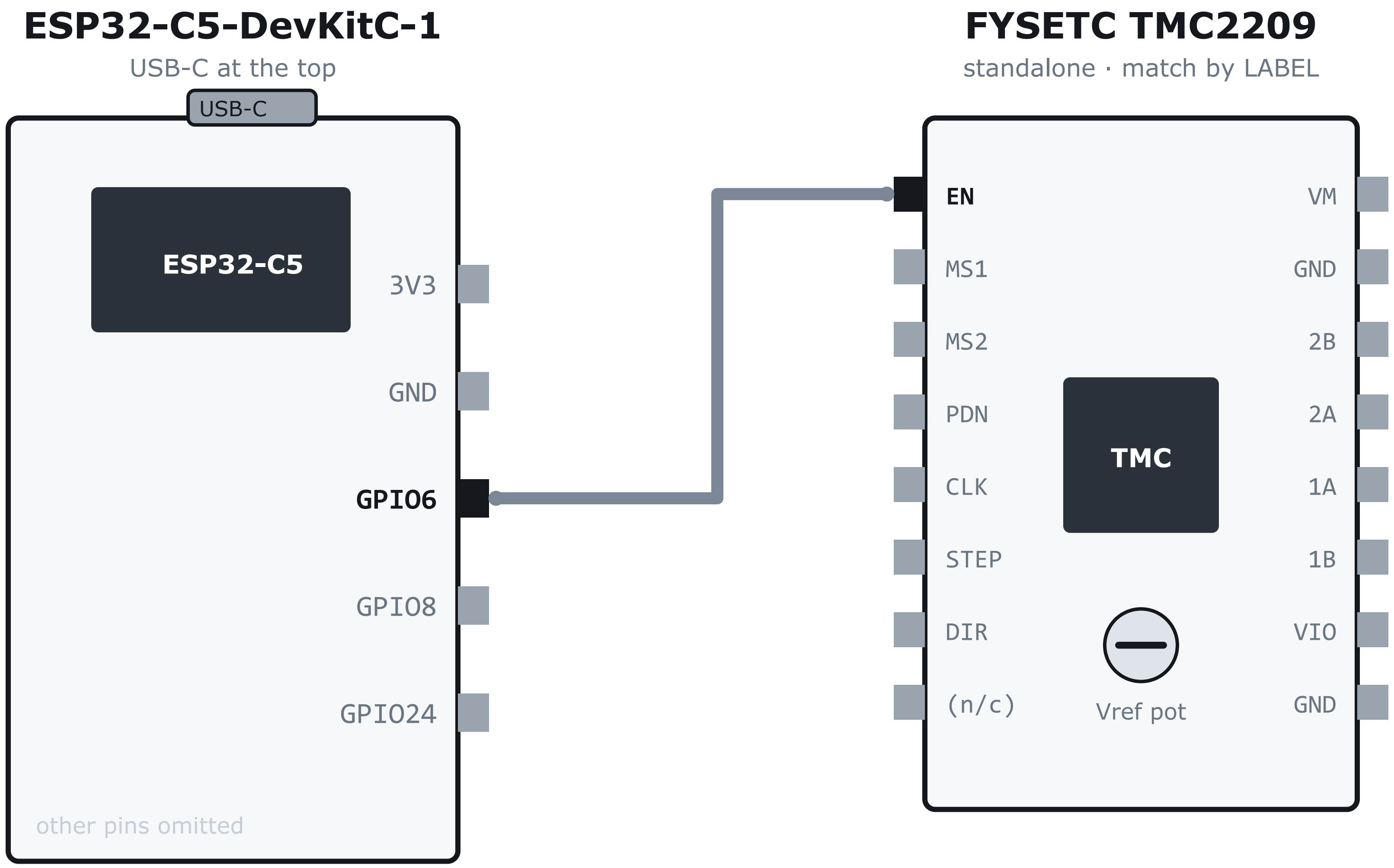
ON THE BOARD

DIR is row 32 column D. Jumper (M-F) row 32 column B -> the ESP pin labelled GPIO24.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

Connection 5 — EN -> GPIO6

wire colour: white / grey



ENABLE, and it is ACTIVE LOW: pulling it low turns the coils on. Most modules pull it high on board, so an unconnected EN means disabled — which is the safe default you want for a first power-on.

CHECK YOURS. With USB in and VM off, measure EN against GND: it should read about 3.3 V. If it reads near 0 V, add a 10 kohm resistor from EN to 3V3 before you ever connect VM, or the coils energise the instant power arrives.

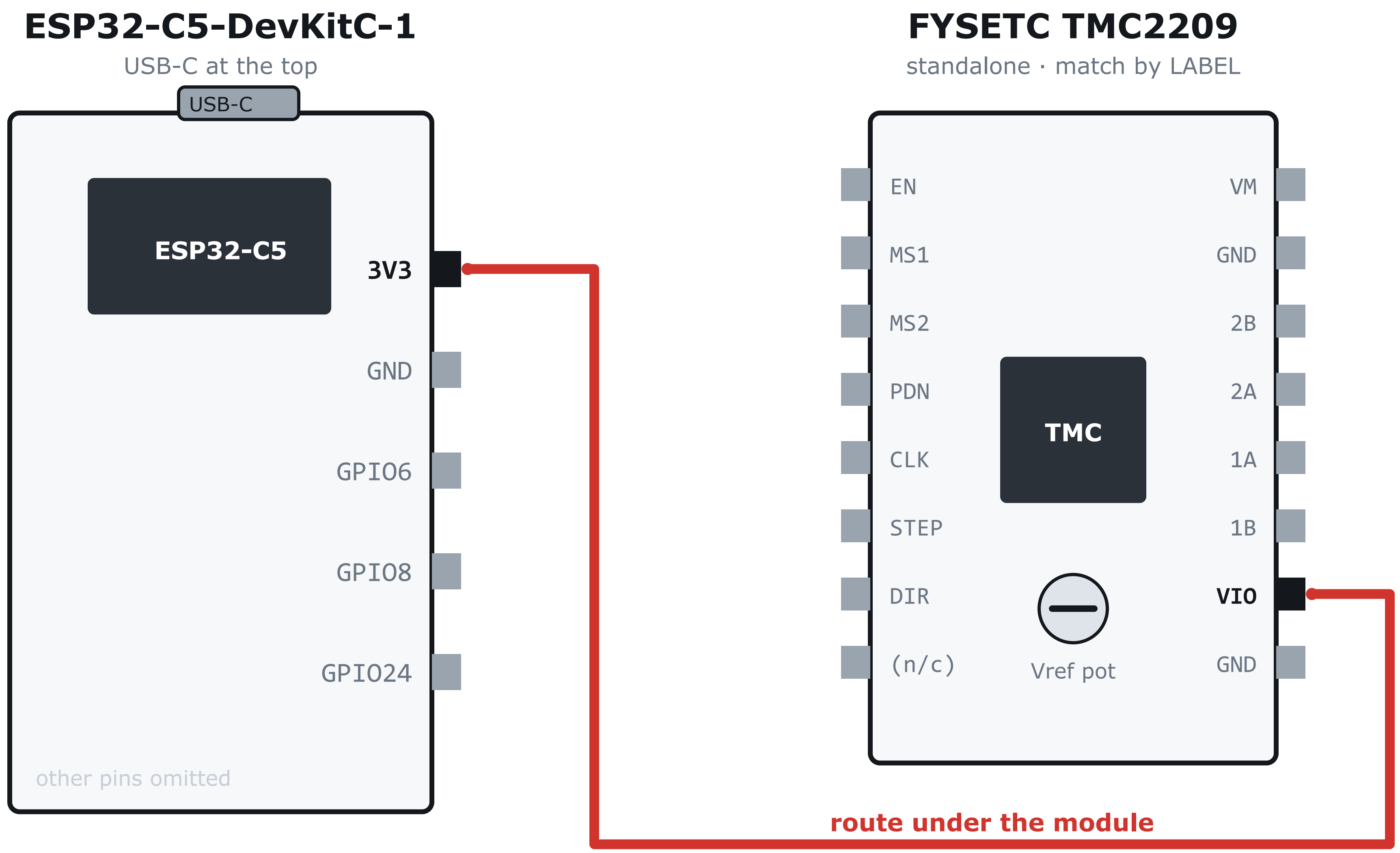
ON THE BOARD

EN is row 26 column D. Jumper (M-F) row 26 column B -> the ESP pin labelled GPIO6.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

Connection 6 — VIO -> 3V3

wire colour: red, thin



The logic supply. It sets the voltage the driver expects to see on STEP, DIR, EN and the MS pins, so it must be the SAME 3.3 V the ESP32 drives with. VIO is on the right-hand pin column, so this one routes under the module.

3.3 V, not 5 V. Feeding VIO 5 V makes the driver want 5 V logic levels from a 3.3 V microcontroller. It appears to work, right up until it does not.

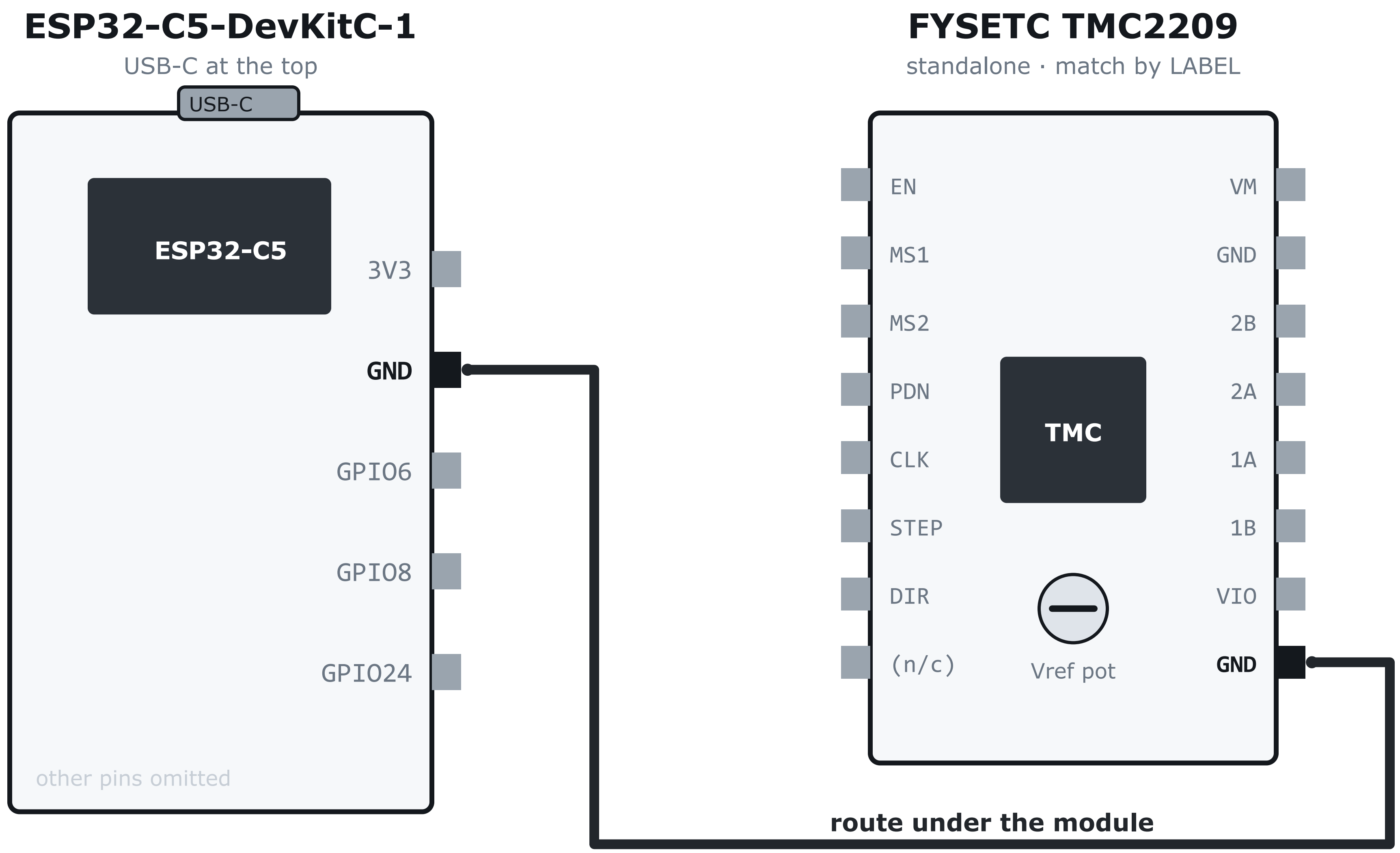
ON THE BOARD

VIO is row 32 column H. Jumper (M-M) row 32 column J -> the + rail.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

Connection 7 — GND (logic) -> GND

wire colour: black, thin



The logic ground. Without a shared ground the ESP32 and the driver have no common reference and every signal level is meaningless. This is the wire whose absence produces the most baffling symptoms.

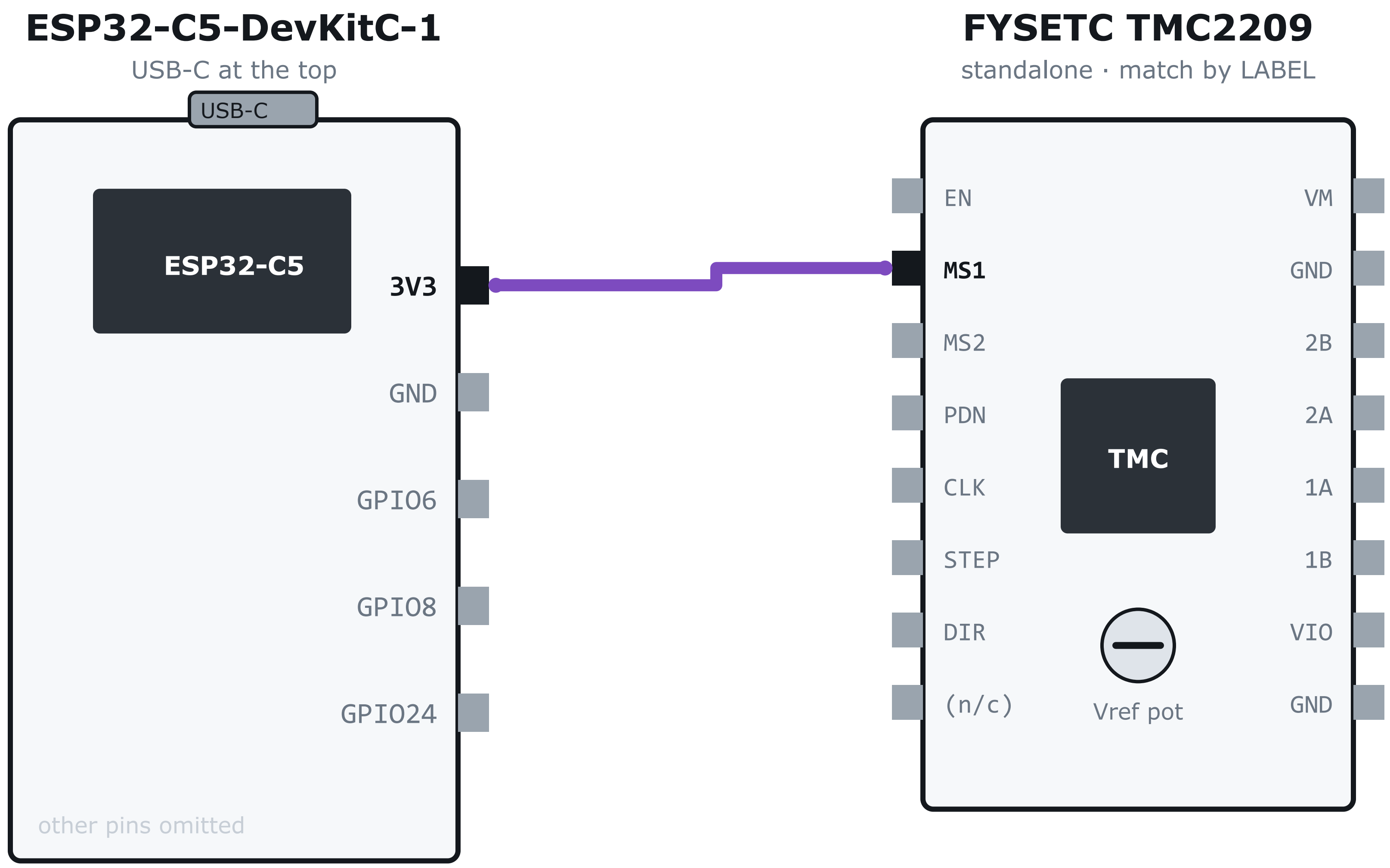
ON THE BOARD

GND is row 33 column H. Jumper (M-M) row 33 column J -> the - rail.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

Connection 10 — MS1 -> 3V3

wire colour: violet



Microstep select, bit one. Tie it HIGH.

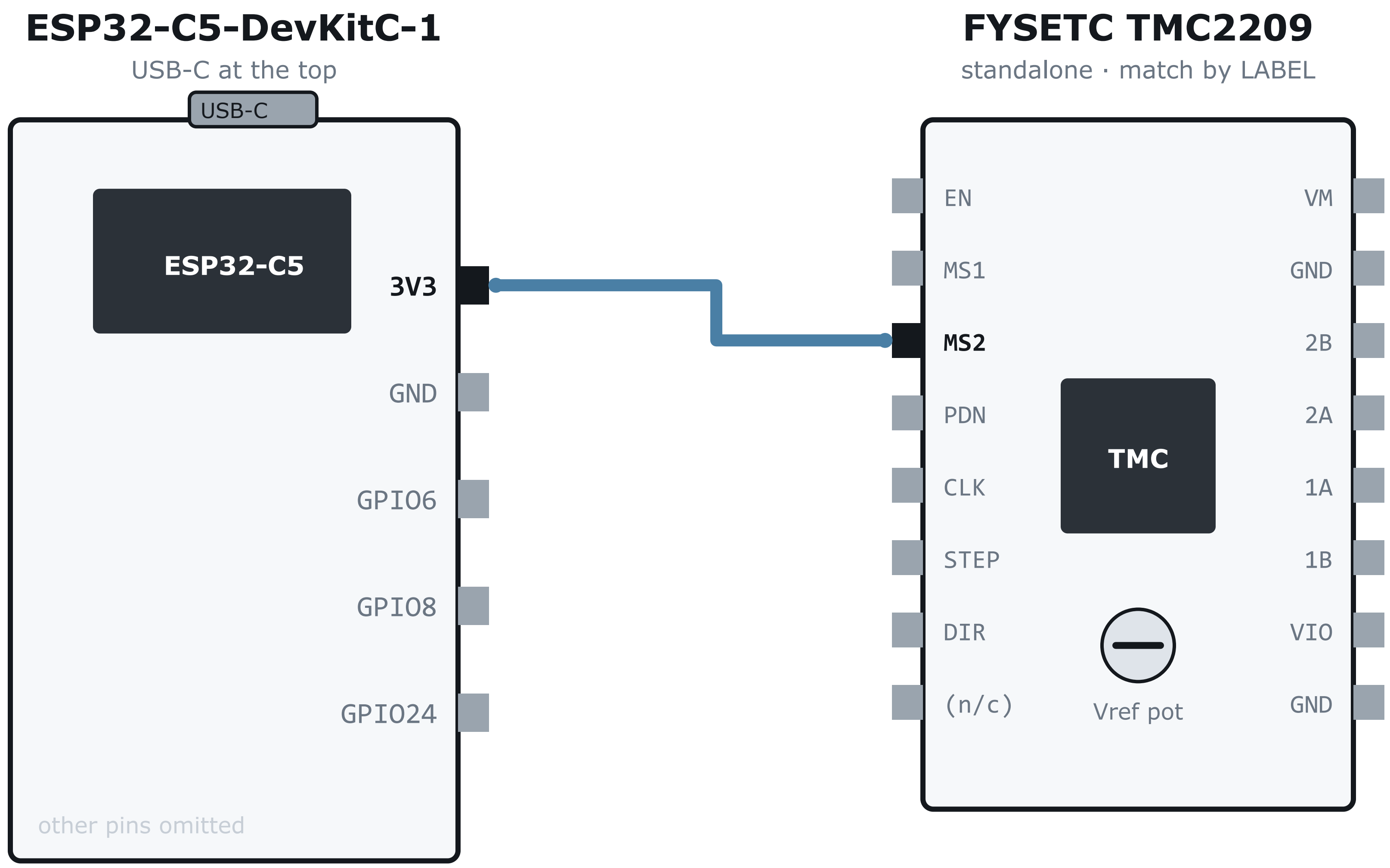
ON THE BOARD

MS1 is row 27 column D. Jumper (M-M) row 27 column B -> the + rail.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

Connection 11 — MS2 -> 3V3

wire colour: grey-blue



TMC2209 microstep table

MS2	MS1	microsteps	
0	0	1/8	
0	1	1/2	
1	0	1/4	
1	**1**	**1/16**	**<- what we want**

Microstep select, bit two. Tie it HIGH as well. BOTH high is 1/16 on a TMC2209, and the table is not in a sensible order — which is exactly why this one is easy to get wrong.

Modules usually pull MS1 and MS2 LOW on board. Leave them unconnected and you get 1/8, which makes every constant in the firmware wrong by a factor of two. Page 26 has a mechanical check that proves the setting — do not trust the jumper.

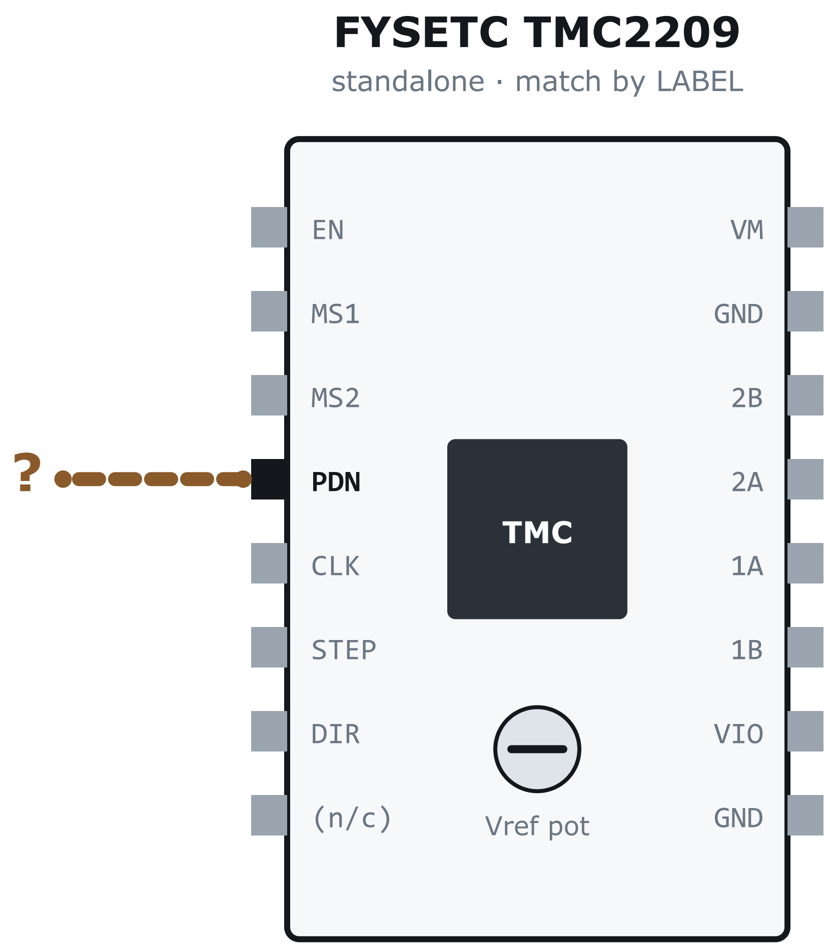
ON THE BOARD

MS2 is row 28 column D. Jumper (M-M) row 28 column B -> the + rail.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

Connection 12 — PDN / UART — standstill current

wire colour: brown



LEAVE AS THE MODULE SHIPS

for standalone operation

then VERIFY on page 28

In standalone mode this pin controls automatic standstill current reduction: about a second after the last step, the driver drops the coil current to roughly half. That is not a nicety here — it is the entire thermal contract, because the motor is sealed in a PLA drum and holds current all day.

THE POLARITY IS NOT VERIFIED for your module and this guide will not guess it. Leave PDN as the module ships for standalone, then confirm on page 28: with the motor energised and still, the supply current must visibly STEP DOWN about a second after the last step. If it never does, tie the pin the other way and re-check.

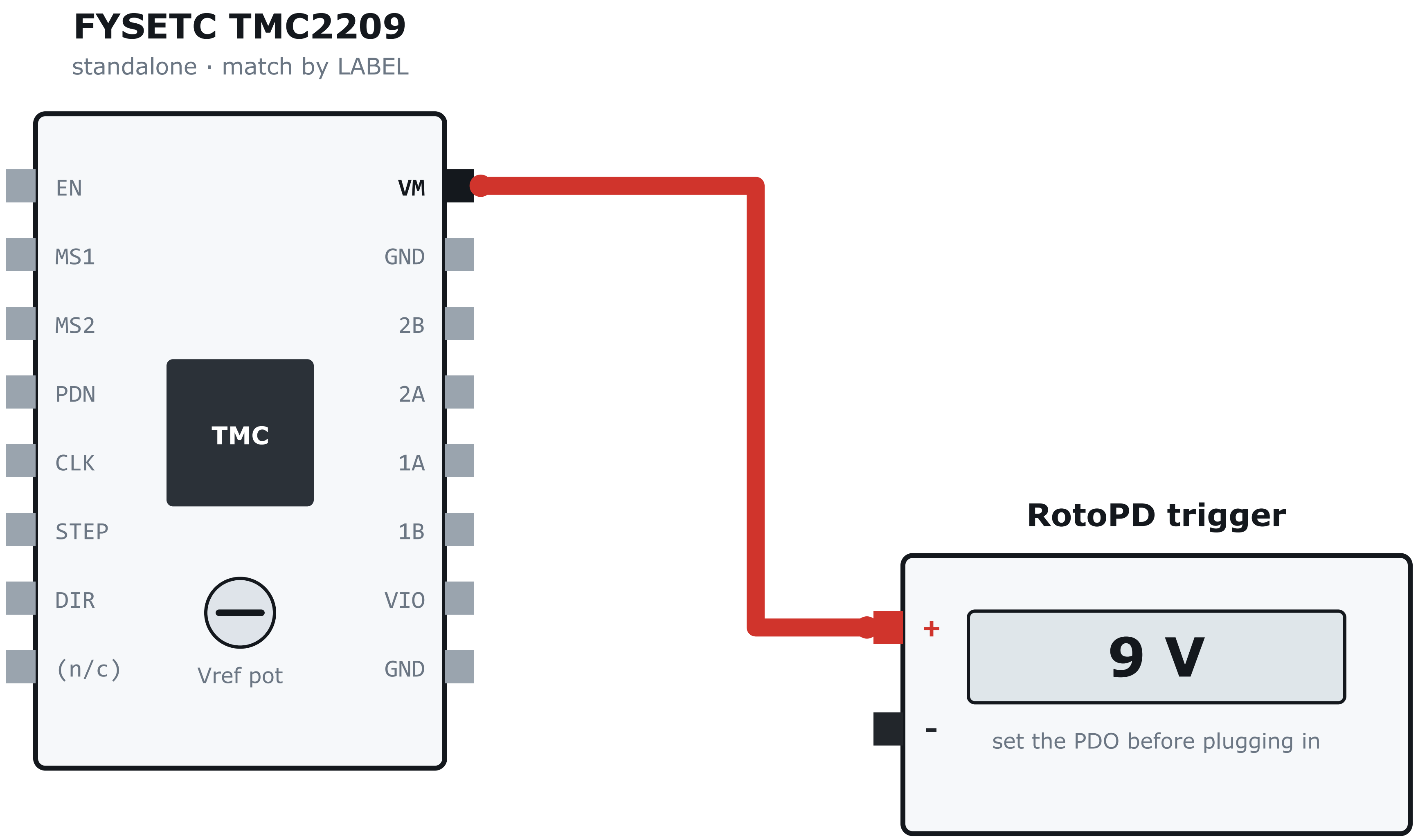
ON THE BOARD

PDN is row 29 column D. NOTHING goes in row 29. Leave the whole row empty.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

Connection 9 — VM -> supply +

wire colour: RED, thick



The motor supply. Start at 9 V from the PD trigger, not 20 V: the TMC2209 runs from 4.75 V up, at one drum revolution per second there is no headroom needed, and a wiring mistake at 9 V dissipates about a fifth of the energy. Move to 20 V once the wiring is proven.

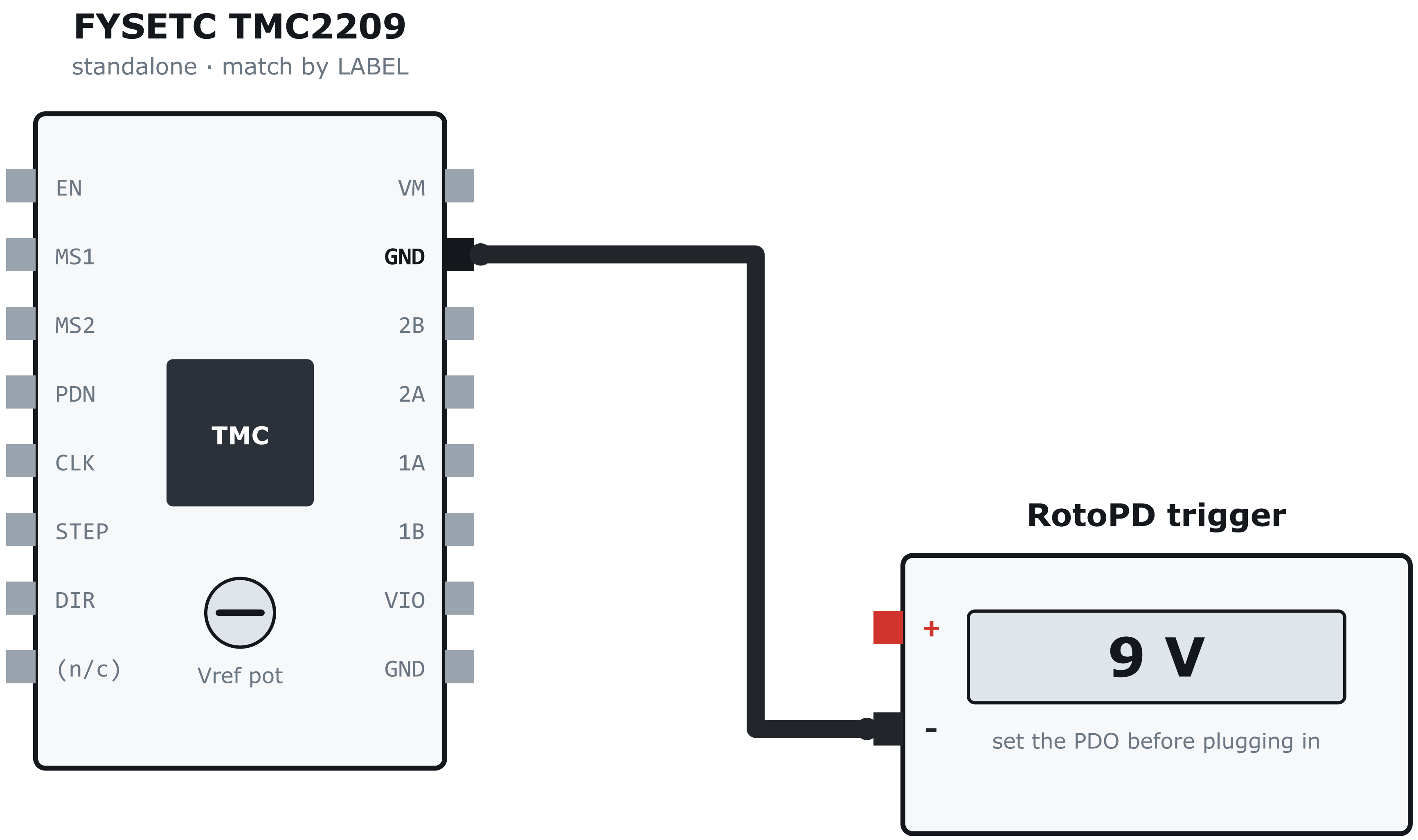
ON THE BOARD

VM is row 26 column H. RotoPD + on its own 22 AWG wire into row 26 column I.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

Connection 8 — GND (power) -> supply -

wire colour: BLACK, thick



The power ground — the wire that carries the coil current back. Keep it short and direct. This is a different job from the logic ground on page 13 even though they are the same net; on a breadboard, route it separately and let them meet at the supply.

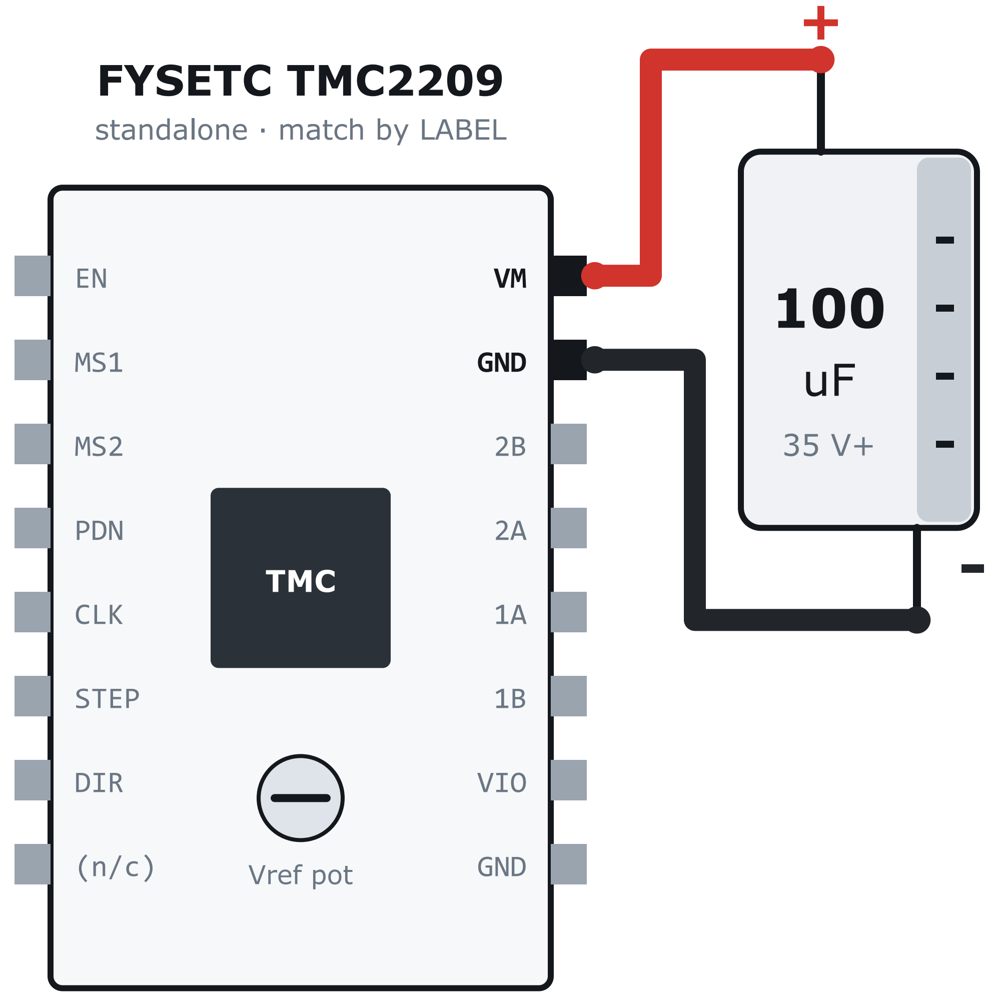
ON THE BOARD

GND is row 27 column H. RotoPD - on its own 22 AWG wire into row 27 column I.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

Bulk capacitor — connection 13

100 uF, at the driver's OWN pins



THE STRIPE IS THE NEGATIVE LEG. Backwards, an electrolytic vents.

35 V minimum; your 50 V part is right. Never a 25 V cap on the 20 V rail — decelerating the drum pushes the rail up.

This capacitor supplies the sudden current the coils demand at each step, which the supply lead is too inductive to deliver. Legs as short as you can make them, bridging the driver's own VM and GND pins.

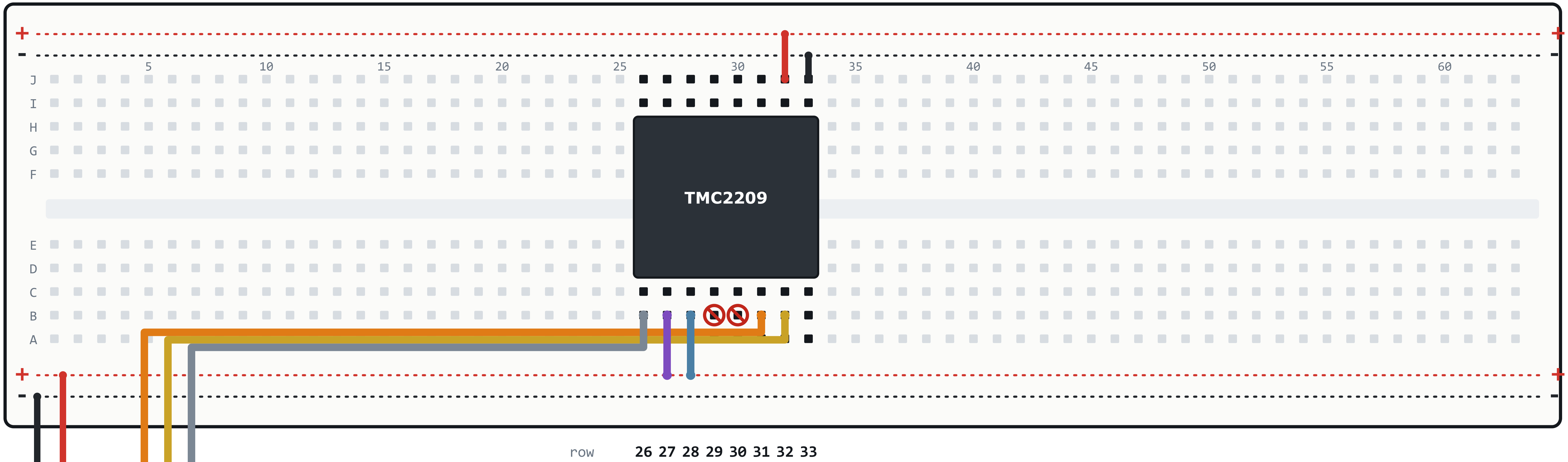
ON A BREADBOARD THIS MEANS THE TWO HOLES BESIDE THE MODULE, not the power rails at the end of the board. Twenty centimetres of breadboard wire defeats the capacitor entirely. This is the single thing breadboard builds get wrong.

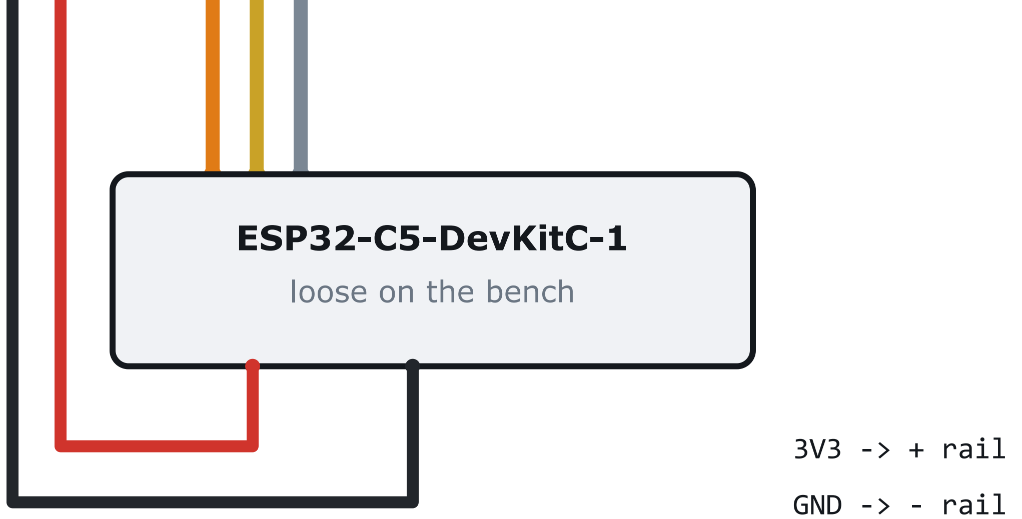
Concretely, on this build: `+ leg row 26 column J - leg row 27 column J`

drawn on page 21; lead prep on page 2

The board as built - logic side

The same connections as pages 9 to 13, by row and column





Every wire that leaves the board for the ESP32 crosses the bottom rails on its way out. That is what a jumper does; it lies on top of them and touches nothing.

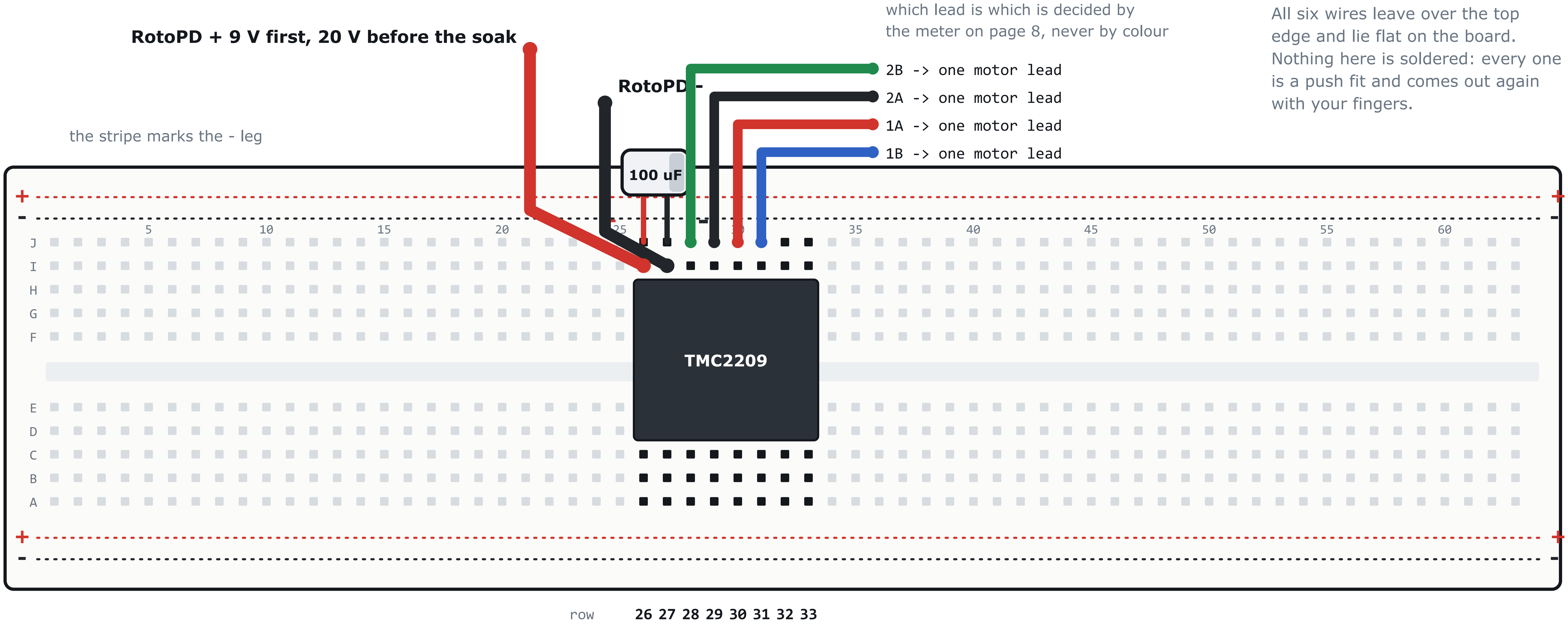
Every wire, by hole

pin	pin is at	wire goes in	and reaches	
STEP	31 D	31 B	ESP GPIO8	M-F
DIR	32 D	32 B	ESP GPIO24	M-F
EN	26 D	26 B	ESP GPIO6	M-F
VIO	32 H	32 J	+ rail	M-M
GND (logic)	33 H	33 J	- rail	M-M
MS1	27 D	27 B	+ rail	M-M
MS2	28 D	28 B	+ rail	M-M
3V3 feed		a rail	+ rail	M-F
GND feed		a rail	- rail	M-F
PDN, CLK	29, 30 B	NOTHING	stay empty	

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

The board as built - power side

The supply, the capacitor and the motor, by row and column



All six wires leave over the top edge and lie flat on the board. Nothing here is soldered: every one is a push fit and comes out again with your fingers.

Every power wire, by hole

what	pin is at	other end	wire goes in	
VM (driver)	26 H	RotoPD +	26 I	22 AWG
GND (power)	27 H	RotoPD -	27 I	22 AWG
100 uF +	-	bridges VM	26 J	legs 12 mm
100 uF -	-	bridges GND	27 J	stripe side
2B (coil)	28 H	pigtail lead	28 J	meter decides
2A (coil)	29 H	pigtail lead	29 J	meter decides
1A (coil)	30 H	pigtail lead	30 J	meter decides
1B (coil)	31 H	pigtail lead	31 J	meter decides

THE CAPACITOR IS NOT OPTIONAL AND ITS POSITION IS THE POINT. Rows 26 and 27 ARE the driver's VM and GND pins, which is why it belongs there and not on a rail twenty centimetres away.

0.7 A RMS needs none of this upgraded - page 2 says why. The coil current never travels along a rail: it runs in the four leads above and inside the driver. The supply wires are separate from the logic ground to keep the switching return out of it, not for current rating.

ROW AND COLUMN NUMBERS ARE LAYOUT, NOT CHECKED FACTS. The pins on this page are parsed from hal/pins.h and BENCH_WIRING.md and cross-checked; no source says which HOLE anything sits in, so nothing verifies that. Build it elsewhere on the board and the electrical pages are still true while these numbers are not.

Continuity check

Beep these five pairs BEFORE the first power-on

Meter on continuity (the beep symbol). Everything unpowered, USB out, motor still disconnected. Five checks: three that must NOT beep, two that MUST. A mistake found here costs a minute; the same mistake found by powering up costs a driver.

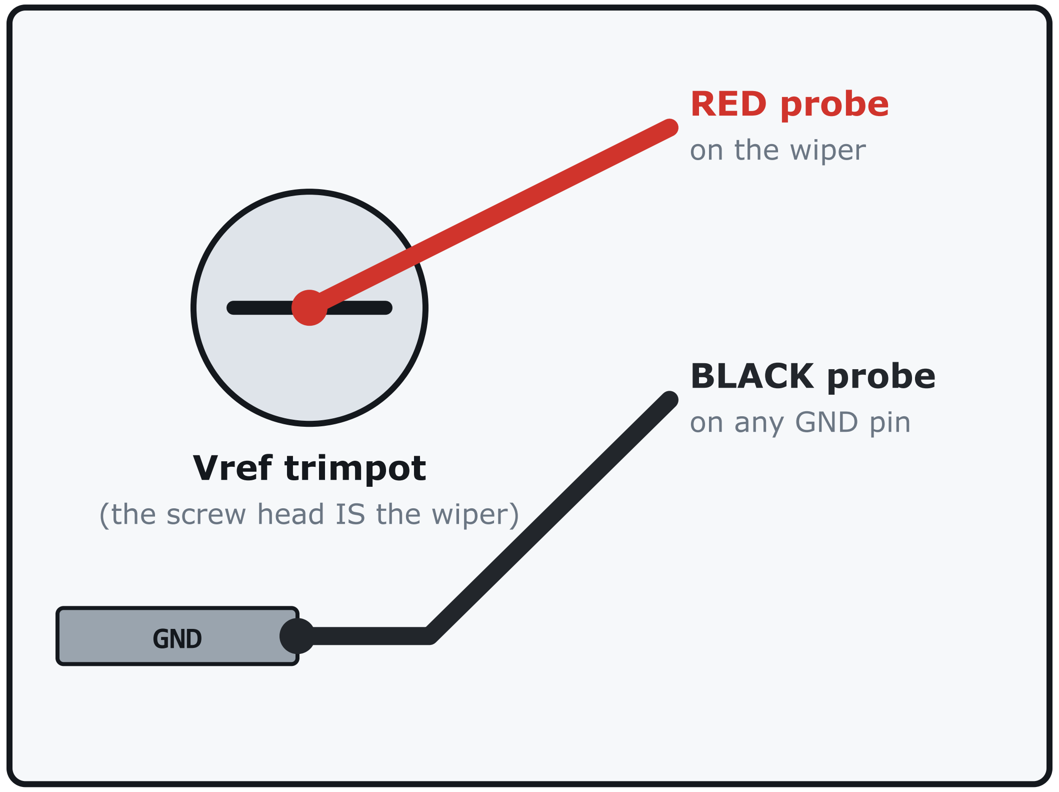
1	`VM <--> GND` **MUST NOT BEEP**	A short across the supply. This is the one that makes smoke.	☐
2	`VM <--> 3V3` **MUST NOT BEEP**	Motor voltage on the ESP32's 3.3 V rail — destroys the board.	☐
3	`3V3 <--> GND` **MUST NOT BEEP**	Shorts the ESP32's regulator; the board will not boot.	☐
4	`ESP GND <--> driver GND` **MUST BEEP**	No common ground means every logic level is undefined. Nothing works, confusingly.	☐
5	`ESP 3V3 <--> driver VIO` **MUST BEEP**	MS1 and MS2 must beep to 3V3 as well — they are on the same net.	☐

All five correct? Then the first power-on cannot destroy anything through a wiring error. Go to page 23.

Set Vref — 0.7 A

Motor disconnected. VM off. USB on.

driver, close up



If your module has a labelled VREF pad, use that instead of the screw. Use a CLIP, not a hand-held probe: one slip from the wiper to a neighbouring pad kills the driver.

1. Read your sense resistor (page 3)

marking	R_sense	full scale	Vref for 0.7 A
R110	0.11 ohm	1.77 A	**0.99 V**
R150	0.15 ohm	1.35 A	**1.29 V**

2. Meter to DC volts, 2 V range. Black on GND.
3. Turn the pot in small steps and read directly.
4. Set it to the figure from the table.

Vref set = ______________________ V

The pot is sensitive — a few degrees is 0.1 V. Creep up on the number; do not sweep past it and come back.

0.99 V is 0.7 A RMS on an R110 module. The A4988 formula would say 0.19 V and give you about 0.14 A — a motor that skips under load and looks too small for the job.

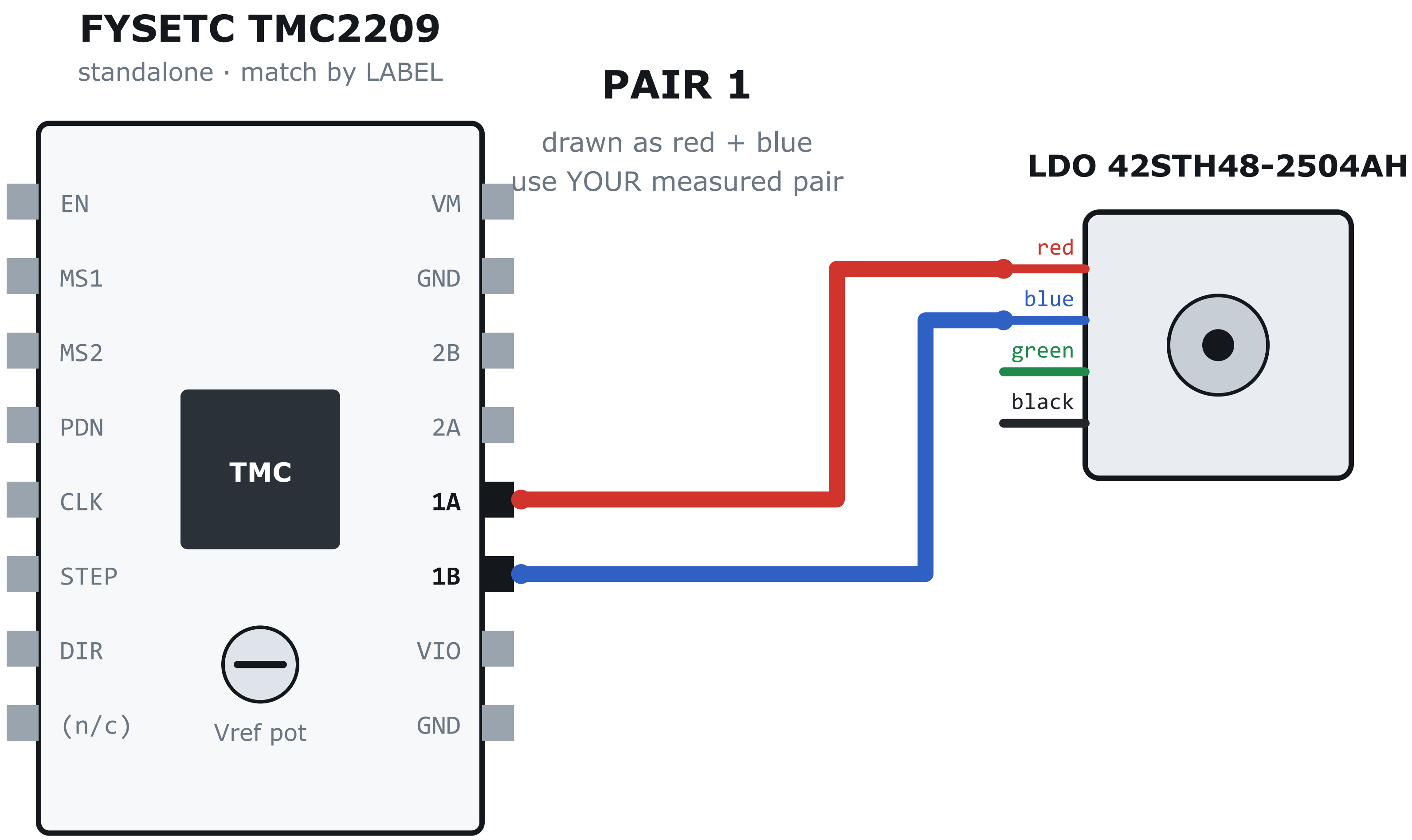
The TMC2209 uses VREF as a SCALING input against a full-scale current fixed by the sense resistor — it is not the A4988 formula, and using that one gives about 0.14 A, at which the motor skips under load and looks inadequate. The equation these numbers come from:

```
I_RMS = [ 0.325 / (R_sense + 0.02) ] / sqrt(2) x ( Vref / 2.5 )
```

MOTOR DISCONNECTED for this whole page. That is rule 3 on page 4: set the current before the motor can be subjected to it.

Connection 1 — coil A

POWER OFF while you do this

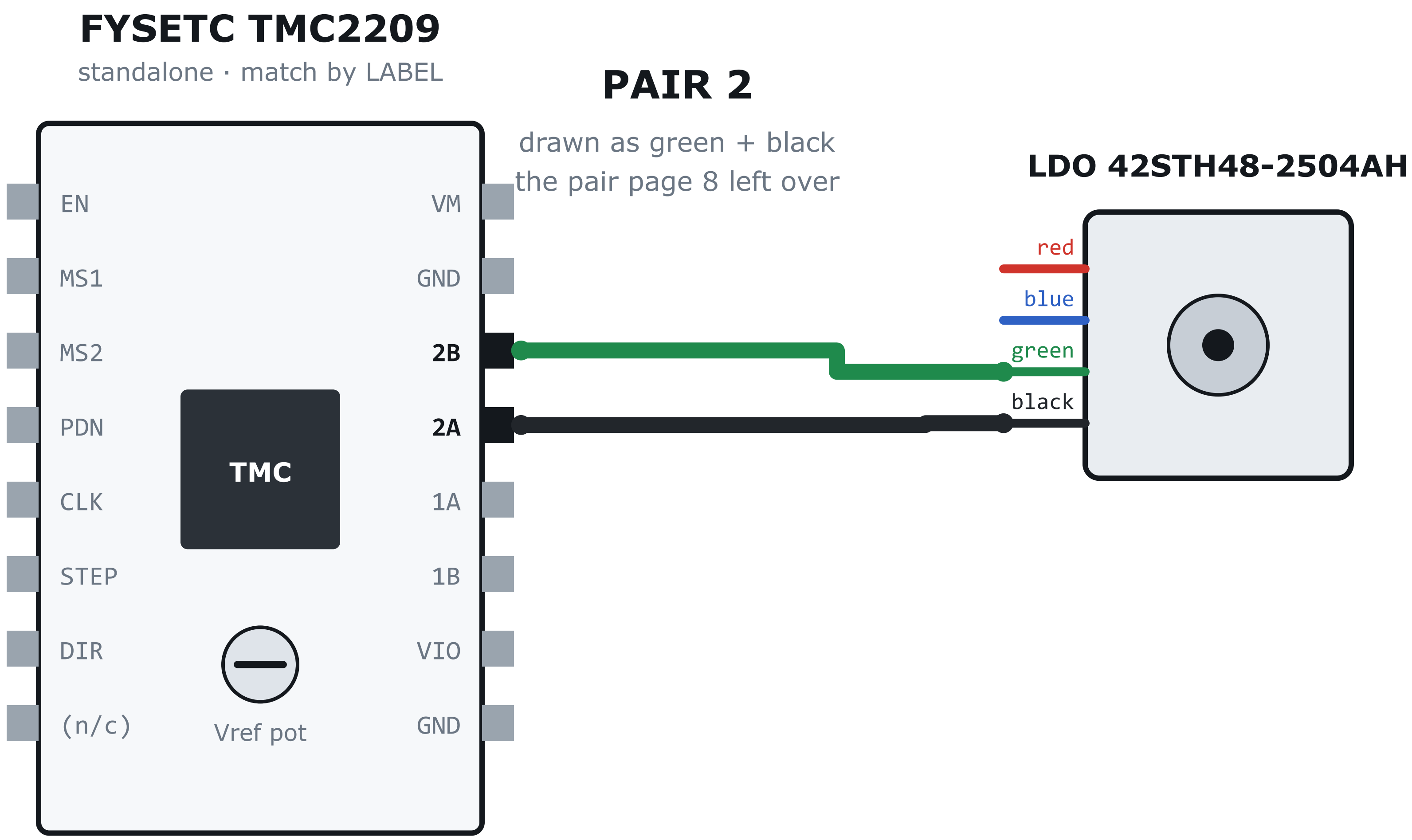


Both ends of ONE coil — the pair you found on page 8 — go to 1A and 1B. Which of the two ends goes to which pin does not matter; it only reverses rotation, and direction is a firmware bit. What matters is that this bridge gets both ends of one coil and never one end of each.

VM MUST BE OFF. Plugging a coil into a live driver is rule 1 on page 4, and it is the most common way these modules die.

Connection 2 — coil B

POWER OFF while you do this

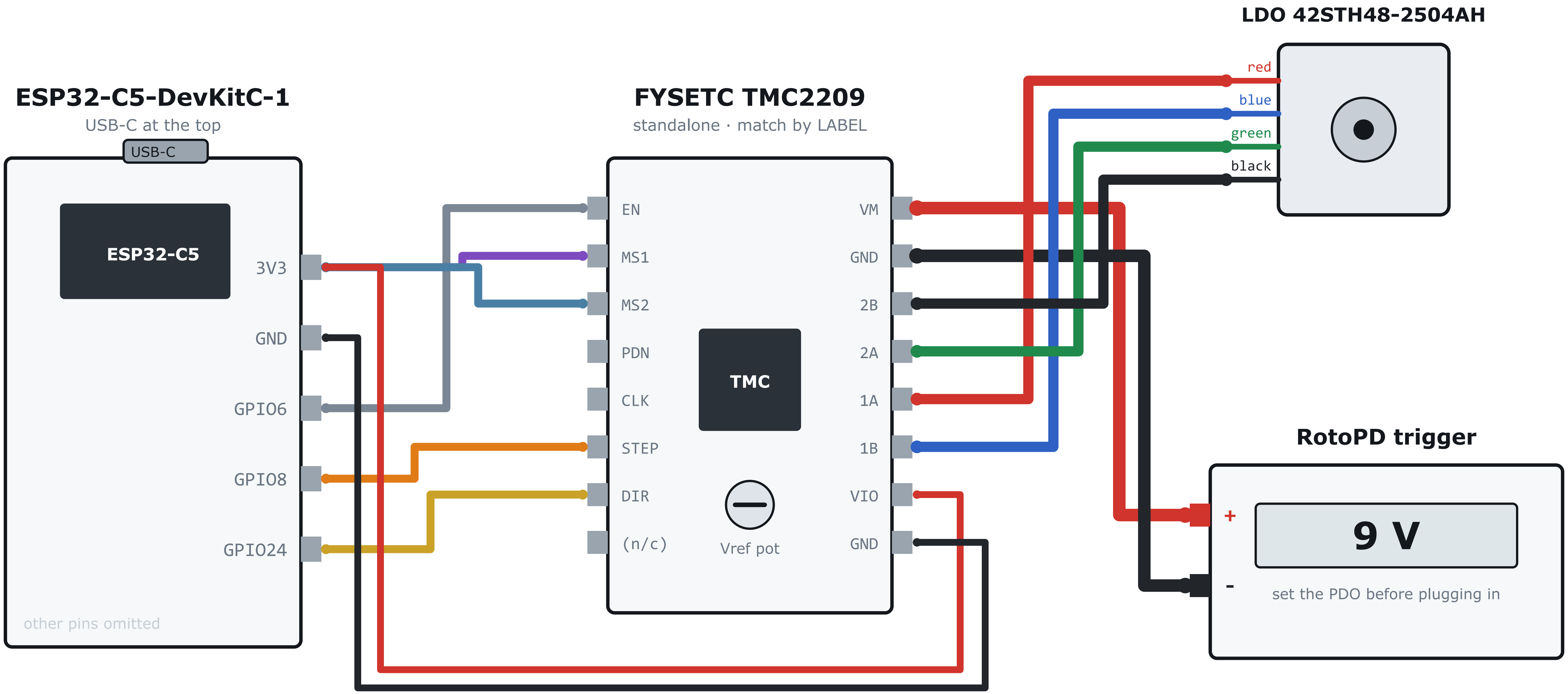


The other coil, to 2A and 2B. Same rule: both ends of one coil into one bridge. Mixing them across the two bridges does not turn the motor — it buzzes, locks or judders, and it stresses the driver.

WITH THE MOTOR CONNECTED AND POWER STILL OFF, one last meter check: 1A to 1B beeps, 2A to 2B beeps, and 1A to 2A does NOT.

Finished layout

Every connection at once - the ELECTRICAL truth. Page 27 is the board itself.

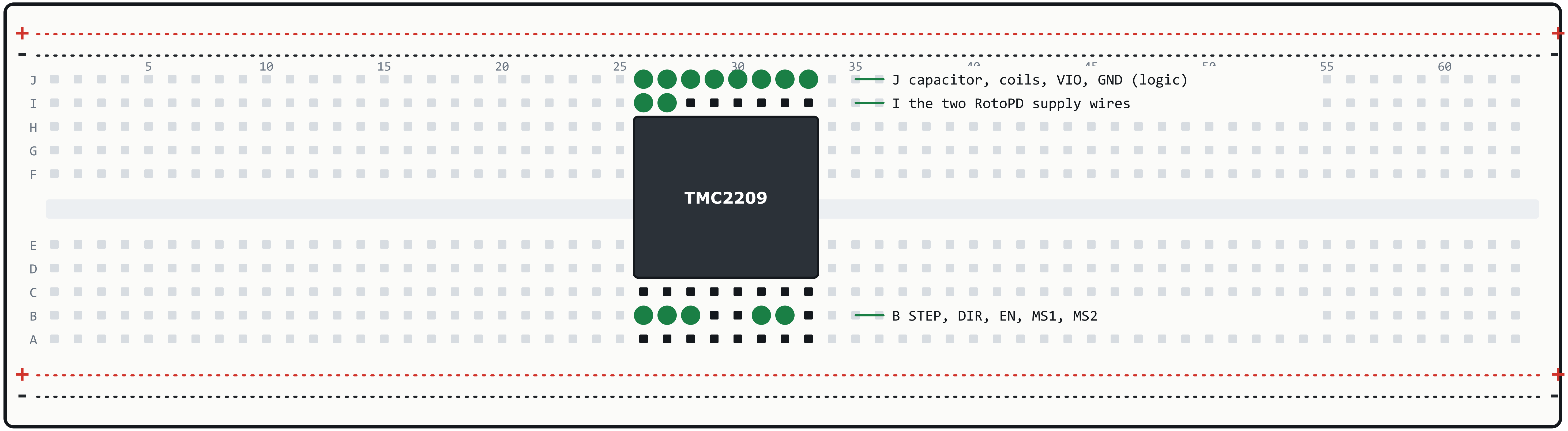


The geometry-proving check, once this is powered:

`step 0 3200`	must be EXACTLY one drum revolution. Two revolutions means MS1/MS2 are giving 1/8; four means 1/4. Judder with no net rotation means the coil pairing is wrong. Mark the drum first.

The finished board

Walk rows 26 to 33 once, before anything is powered



The whole board, row by row

☐	row 26	EN | VM	B EN, I VM in, J cap +
☐	row 27	MS1 | GND	B MS1, I GND in, J cap -
☐	row 28	MS2 | 2B	B MS2, J 2B
☐	row 29	PDN | 2A	J 2A
☐	row 30	CLK | 1A	J 1A
☐	row 31	STEP | 1B	B STEP, J 1B
☐	row 32	DIR | VIO	B DIR, J VIO
☐	row 33	(n/c) | GND	J GND (logic)

Count before you power anything

9	jumper wires
2	supply wires from the RotoPD
2	capacitor legs
4	motor leads from the pigtail
17	**things in the board, total**

A hole you cannot account for is a wire in the wrong row. Pull it out and find it on page 20 or 21.

Then the meter, page 22. It finds what a count cannot: two wires in one row that should not share it.

Power on, and power off

In this order, every time

POWER ON

1. Everything wired, MOTOR INCLUDED, nothing powered.
2. USB-C from the PC to the ESP32. VIO comes up; logic defined.
3. Confirm EN reads high (disabled) — page 11. USB in, VM still off.
4. Watch the console boot. Board healthy BEFORE any motor voltage.
5. maint on — stops it hunting for a Hall that is not there.
6. Apply VM (9 V). Output stage now live.
7. en 1 — and only now are the coils energised.

STOP NOW — cut VM immediately

- any smell of hot plastic or varnish
- the driver too hot to keep a finger on
- the motor case climbing past uncomfortable early in a run
- the bulk capacitor warm, or bulging
- loud buzzing with no rotation (coil pairing — page 8)
- supply current spiking, or the supply cutting out
- the ESP32 rebooting repeatedly (brownout — check grounds)

Do not go looking for the console if something smells hot. Pull VM.

POWER OFF

1. en 0 — de-energise the coils first.
2. Remove VM.
3. Remove USB.
4. Before touching the motor plug: confirm the bulk cap has discharged.

THE BULK CAPACITOR HOLDS CHARGE after the supply is gone. That is still a live driver as far as rule 1 is concerned.

Standstill-current check (page 16)

With the motor energised and stationary, watch the supply current: it must STEP DOWN about a second after the last step. That is standstill reduction working, and it is the whole thermal contract. If it never steps down, PDN is tied the wrong way.

EXPECTED, not a problem: faults on columns 1-4 (nothing is wired to them), a no_hall fault on column 0 before `maint on`, a steady hiss at standstill (the chopper), and the drum settling to a slightly different rest position after `en 0` — that is the 3.92 N·cm imbalance against a 2.2 N·cm detent, and seeing it confirms the premise.